

GATE MOCK PAPER -3 2026
Data Science & Artificial Intelligence (DA)

General Aptitude

Q.1 – Q.5 Carry ONE mark Each

Q.1	If “ $\rightarrow$ ” denotes increasing order of abstraction, then Concrete $\rightarrow$ Generic $\rightarrow$ Abstract is analogous to: Letter $\rightarrow$ _____ $\rightarrow$ Language
(A)	Word
(B)	Sentence
(C)	Alphabet
(D)	Grammar

Q.2	Some problems appear unsolvable until constraints are relaxed. Once constraints are modified, feasible solutions may emerge. Which statement must be TRUE?
(A)	All constraints hinder solutions
(B)	Constraint relaxation can lead to feasibility
(C)	Original constraints are incorrect
(D)	Relaxed problems are always optimal

Q.3	A box contains 4 red, 3 blue, and 2 green balls. Two balls are drawn without replacement. What is the probability that both balls are of different colors?
(A)	$\frac{19}{36}$
(B)	$\frac{5}{9}$
(C)	$\frac{7}{9}$
(D)	$\frac{17}{36}$

Q.4	A regular hexagon is inscribed in a circle of radius r . The ratio of the area of the hexagon to the area of the circle is
(A)	$\frac{3\sqrt{3}}{2\pi}$
(B)	$\frac{6}{\pi}$
(C)	$\frac{\sqrt{3}}{\pi}$
(D)	$\frac{2\sqrt{3}}{\pi}$

Q.5	The mean of n numbers is 50. If each number is increased by its index position, the new mean becomes 53. The value of n is
(A)	4
(B)	5
(C)	6
(D)	7

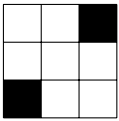
Q.6 – Q.10 Carry TWO marks Each

Q.6	Let $f(x) = x^4 - 4x^2 + 1$. Which statements are correct?
(A)	$f(x)$ is an even function
(B)	$f(x)$ has two local minima
(C)	$f(x)$ has one local maximum
(D)	Global minimum is -3

Q.7	Let events A, B satisfy $P(A) = 0.5$, $P(B) = 0.4$, $P(A B) = 0.6$. Which are TRUE?
(A)	$P(A \cap B) = 0.24$
(B)	A and B are independent
(C)	$P(B A) = 0.48$
(D)	$P(A \cup B) = 0.66$

Q.8	Consider the series $\sum_{n=2}^{\infty} \frac{1}{n \ln n}.$ Which statements are correct?
(A)	Diverges by integral test
(B)	Converges conditionally
(C)	Partial sums grow slowly
(D)	Comparison with harmonic series applies

Q.9	A 5×5 grid starts with the central cross shaded. A cell becomes shaded if it has an odd number of shaded neighbors. Final number of shaded cells is
(A)	5
(B)	9
(C)	13
(D)	25

Q.10	Which option completes the pattern? 
(A)	Symmetric fill
(B)	Diagonal fill
(C)	Center fill
(D)	Random fill

Q.11 – Q.35 Carry ONE mark Each

Q.11	If X, Y are independent with finite variance, which is always true?
(A)	$Var(X + Y) = Var(X) + Var(Y)$
(B)	$E[XY] = 0$
(C)	X, Y must be normal
(D)	$Cov(X, Y) \neq 0$
Q.12	Three fair coins are tossed. Let A be the event that at least two heads occur and B be the event that the first toss is a tail. What is $P(A B)$?
(A)	$\frac{1}{4}$
(B)	$\frac{1}{3}$
(C)	$\frac{1}{2}$
(D)	$\frac{2}{3}$

Q.13	A 3×3 matrix has eigenvalues 0, 2, 5. Which statement is TRUE?
(A)	Matrix is invertible
(B)	Rank is 2
(C)	Trace is 7
(D)	Determinant is 10

Q.14	In DFS of a directed graph, which edge type indicates a cycle?
(A)	Tree edge
(B)	Back edge
(C)	Forward edge
(D)	Cross edge

Q.15	If $f'(x_0) = 0$ and $f''(x_0) = 0$, which conclusion is valid?
(A)	Local maximum
(B)	Local minimum
(C)	Inflection possible
(D)	Global extremum

Q.16	Which data structure supports both stack and queue operations efficiently?
(A)	Priority queue
(B)	Deque
(C)	Hash table
(D)	Binary heap

Q.17	Which factor controls margin width in soft-margin SVM?
(A)	Kernel
(B)	Learning rate
(C)	Regularization parameter C
(D)	Bias term

Q.18	Which property is maximized by PCA?
(A)	Class separation
(B)	Variance
(C)	Likelihood
(D)	Margin

Q.19	In k -means, which situation causes convergence to poor local minima?
(A)	Large dataset
(B)	Bad initialization
(C)	High dimensions
(D)	Using Euclidean distance

Q.20	Which regularization encourages sparsity in weights?
(A)	Ridge
(B)	Lasso
(C)	Elastic Net
(D)	Tikhonov

Q.21	Recurrence $T(n) = T(n - 1) + n$ solves to
(A)	$O(n)$
(B)	$O(n \log n)$
(C)	$O(n^2)$
(D)	$O(2^n)$

Q.22	Why does Naïve Bayes perform well in high dimensions?
(A)	Assumes independence
(B)	Uses distance metrics
(C)	Reduces variance
(D)	Avoids curse of dimensionality

Q.23	Which formula is a tautology?
(A)	$p \wedge \neg p$
(B)	$p \vee \neg p$
(C)	$p \rightarrow \neg p$
(D)	$\neg(p \vee q)$

Q.24	If $X \sim \text{Exp}(\lambda)$, then $E[X^2]$ equals
(A)	$\frac{1}{\lambda^2}$
(B)	$\frac{2}{\lambda^2}$
(C)	$\frac{1}{\lambda}$
(D)	$\frac{2}{\lambda}$

Q.25	Which problem is NP-complete?
(A)	Clique
(B)	Topological sort
(C)	MST
(D)	BFS

Q.36 – Q.65 Carry TWO marks Each

Q.36	Let $X_1, \dots, X_{200}$ be i.i.d. with mean 2 and variance 4. Using CLT, approximate $P(360 \leq \sum X_i \leq 440)$.
(A)	$\Phi(1) - \Phi(-1)$
(B)	$\Phi(2) - \Phi(-2)$
(C)	$\Phi(0.5) - \Phi(-0.5)$
(D)	$\Phi(3) - \Phi(-3)$

Q.37	Which estimators are consistent?
(A)	Sample mean
(B)	Sample variance
(C)	Sample median
(D)	Maximum observation

Q.38	In Bayesian network $A \rightarrow B \leftarrow C$, observing B makes
(A)	A independent of C
(B)	A dependent on C
(C)	B independent
(D)	No change

Q.39	Which statement about SVD is TRUE?
(A)	Singular values are eigenvalues
(B)	Rank equals number of nonzero singular values
(C)	U, V are arbitrary
(D)	Only for square matrices

Q.40	Which properties hold for orthogonal matrices?
(A)	$A^T A = I$
(B)	Preserves dot product
(C)	Eigenvalues magnitude 1
(D)	Always symmetric

Q.41	Consider the function $f(x) = x^4 - 2x^2 + 1.$ Which of the following statements are TRUE?
(A)	$f(x)$ has exactly two global minima
(B)	$f(x)$ is convex for all x
(C)	$x = 0$ is a local maximum
(D)	$f(x)$ has exactly three stationary points

Q.42	Which of the following statements about relational keys are TRUE?
(A)	Every candidate key is a superkey
(B)	A superkey is always minimal
(C)	A relation may have multiple candidate keys
(D)	Prime attributes must appear in all candidate keys

Q.43	Which of the following are TRUE for decision trees trained to zero error?
(A)	Training accuracy is 100%
(B)	Test error is necessarily low
(C)	Overfitting is possible
(D)	Pruning can reduce variance

Q.44	Which SQL queries are syntactically and semantically valid?
(A)	SELECT dept FROM Emp GROUP BY dept HAVING COUNT(*) > 1;
(B)	SELECT dept, salary FROM Emp GROUP BY dept;
(C)	SELECT COUNT(*) FROM Emp WHERE salary < 50000;
(D)	SELECT dept FROM Emp WHERE COUNT(*) > 2;

Q.45	Which problems admit optimal greedy solutions?
(A)	Interval scheduling
(B)	Minimum spanning tree
(C)	0/1 knapsack
(D)	Fractional knapsack

Q.46	Which properties of heuristic $h(n)$ are sufficient for optimal A* search?
(A)	Admissibility
(B)	Consistency
(C)	$h(n) = 0$ for all n
(D)	$h(n) \geq 0$

Q.47	Which statements about Bellman–Ford are TRUE?
(A)	Time complexity is $O(VE)$
(B)	Works with negative edges
(C)	Fails if no negative cycle exists
(D)	Can detect negative cycles

Q.48	A relation has 8 attributes and exactly two candidate keys of size 3 each. The number of prime attributes is _____

Q.49	A connected graph with n vertices and $n - 1$ edges is a tree. Minimum number of edges to be added to make it 2-edge-connected is _____

Q.50	Let X and Y be independent random variables with finite variance. Which are TRUE?
(A)	$Cov(X, Y) = 0$
(B)	$E[XY] = E[X]E[Y]$
(C)	$X + Y$ is independent of X
(D)	$Var(X + Y) = Var(X) + Var(Y)$

Q.51	Let A be an orthogonal matrix with $\det(A) = -1$. Which statements are TRUE?
(A)	$A^{-1} = A^T$
(B)	A represents a reflection
(C)	All eigenvalues have magnitude 1
(D)	A must be symmetric

Q.52	Which statements regarding normalization are TRUE?
(A)	BCNF implies 3NF
(B)	3NF implies BCNF
(C)	BCNF may lose dependencies
(D)	3NF guarantees dependency preservation

Q.53	Which properties hold for B^+ -trees?
(A)	All data entries appear at leaf nodes
(B)	Leaf nodes are linked
(C)	Height is $O(\log n)$
(D)	Internal nodes store full records

Q.54	For a graph with negative edge weights but no negative cycle:
(A)	Dijkstra's algorithm fails
(B)	Bellman–Ford computes correct shortest paths
(C)	Negative edge implies negative cycle
(D)	Bellman–Ford can detect cycles if present

Q.55	A standard deck of 52 cards is dealt equally among 4 players. The probability that each player gets exactly one ace is
(A)	$\frac{4! \binom{48}{12}^4}{\binom{52}{13}^4}$
(B)	$\frac{\binom{4}{1}^4}{\binom{52}{13}}$
(C)	$\frac{1}{13^4}$
(D)	$\frac{4!}{52!}$

Q.56	Let X_n be Bernoulli with $P(X_n = 1) = \frac{1}{n}$. Which modes of convergence to 0 hold?
(A)	In probability
(B)	In mean
(C)	Almost surely
(D)	In distribution

Q.57	Which are advantages of Gini impurity over entropy?
(A)	Faster computation
(B)	Less sensitive to class imbalance
(C)	Produces balanced splits
(D)	Always higher accuracy

Q.58	A matrix has eigenvalues 1, 2, 4. Find $\text{trace}(A^{-1})$. _____

Q.59	Which statement about alpha-beta pruning is TRUE?
(A)	It changes minimax value
(B)	It prunes branches that cannot affect decision
(C)	Worst-case complexity equals minimax
(D)	It applies only to deterministic games

Q.60	For $f(x) = \begin{cases} \frac{\sin(ax)}{x}, & x \neq 0 \\ b, & x = 0 \end{cases}$ Which are TRUE?
(A)	Continuous at 0 if $b = a$
(B)	Differentiable at 0 if $a \neq 0$
(C)	$f'(0) = 0$ when $a = 1$
(D)	Not differentiable if $a \neq 0$

Q.61	A binary min-heap initially contains 1023 elements. Maximum number of key comparisons during one Insert followed by one Extract-Min is _____

Q.62	A binary heap has 18 elements. Insert 7 elements and perform 5 Extract-Min operations. Final heap size is _____

Q.63	A full binary game tree of depth 6 is searched using minimax without pruning. Number of leaf nodes evaluated is _____

Q.64	Relation $R(A, B, C, D, E)$ has FDs: $A \rightarrow B, B \rightarrow C, CD \rightarrow E$. Number of candidate keys is _____

Q.65	A B⁺-tree of order 4 indexes 2048 keys. Minimum possible height of the tree is _____